

Viability-Preserving Rollout Approximate Dynamic Programming for Required-Target Orbital Inspection

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Abstract— Autonomous orbital inspection must complete prescribed observations while respecting relative-motion and safety constraints. This paper formulates required-target inspection as rollout approximate dynamic programming (ADP) on a directed graph of audited orbital arcs. A finite Bellman prefix uses the complete cost of a deterministic task-aware base policy as its approximate terminal value; an uncertified completion has infinite value. The augmented state retains covered targets, selected viewpoints, and residual budget. Under the deterministic finite-graph assumptions, exhaustive rollout preserves a finite completion certificate and cannot exceed the cost of its completing base policy. Required-target completion is distinguished from optional background coverage, and safe connecting actions need not reveal a new target. Geometry-defined requirements on an International Space Station mesh are frozen independently of scenario outcomes.

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Supplementary material containing data and code is available with this article. Color versions of one or more figures are available online at <https://ieeexplore.ieee.org>.

A fresh nine-target confirmation retains all 50 test and 30 shifted scenarios, including loss of required visibility. Depth-three ADP completes 43 and 9 missions, respectively, with all nine targets credited in each successful mission. Paired cost comparisons use only jointly complete routes and a declared one-step-ADP-seeded local-search comparator. On 43 jointly complete test pairs, mean maneuver velocity increment decreases by 11.05%, with a paired reduction of 1.744 m/s (95% interval: 1.525–1.923 m/s). Target-identity accounting is verified separately from historical Robot Operating System 2 execution evidence. The contribution is model-based ADP; the study does not claim full-station inspection, trained deep reinforcement learning, or flight validation.

Index Terms— spacecraft inspection, approximate dynamic programming, safe observable orbital arc, relative orbital dynamics, inspection requirements

I. Introduction

Inspection of large orbital structures is an enabling task for on-orbit servicing and proximity operations. Their external surfaces may contain configuration changes, impact damage, degraded thermal protection, or docking-interface anomalies that fixed onboard cameras cannot observe from a useful geometry [1]–[3]. An autonomous inspector must therefore decide both where to look and how to move. These decisions are tightly coupled in orbit because a camera pose with high geometric coverage can be unreachable, control-intensive, or unsafe under relative motion.

Existing methods address important parts of this coupling at different resolutions. Robotic next-best-view methods treat sensor range, occlusion, incidence angle, and marginal coverage [4]; kinodynamic coverage planners integrate observation and differential constraints in continuous state space [1], [2]; and relative-orbit planners exploit natural orbital motion [5], [6]. A prescribed inspection task additionally distinguishes required surface targets from optional background coverage. An aggregate area or priority threshold can be satisfied while a required target remains unobserved. We therefore prescribe target identities and require their complete inspection, while allowing the planner to choose the camera viewpoints and their order. A view-first pipeline that checks transfers only after selection can still overestimate useful coverage: individually attractive views may be connected only by motions that violate input or keep-out limits.

The sequential challenge is to preserve a feasible completion of every required observation while improving motion cost. A static viewpoint contains no evidence that the spacecraft can reach that pose under its relative dynamics. Conversely, a safe transfer can be useful as a connection even when it reveals no new target immediately. We represent an action by its transfer, stabilized camera observation, and admission audits. Approximate dynamic programming then operates on this finite representation, with a terminal value that certifies completion of the prescribed task.

The safe observable orbital arc (SOOA) is a source-dependent finite-duration Hill–Clohessy–Wiltshire (HCW) transfer from the current graph node to a stabi-

lized candidate observation. Each SOOA references its destination's camera-valid target set and stores control effort, minimum sampled clearance, terminal error, input and speed status, and optional passive-drift audit. Geometric viewpoints define graph nodes, but they do not contribute inspection credit until the corresponding directed SOOA has passed the enabled feasibility tests. Fig. 1 summarizes the progression from mesh targets and camera visibility to a directed graph of admissible inspection actions. The graph objective minimizes accumulated stage cost subject to complete required-target inspection; optional surface coverage is reported separately.

The main contribution is viability-preserving rollout approximate dynamic programming (ADP). A node identifier alone is not Markov because the remaining task depends on covered targets, selected observations, and residual budget. A deterministic task-aware base policy supplies an approximate cost-to-go by completing the remaining mission; finite-depth Bellman expansion improves the first action, the state is updated, and the process repeats [7]. Assigning infinite value to uncertified completions prevents the approximate value from trading away a required target. The resulting policy preserves an available completion and does not increase its base policy's graph cost under the stated deterministic assumptions. Exact dynamic programming supplies a reference on graphs small enough to exhaust.

The contributions are:

- 1) a rollout ADP formulation that combines a required-target terminal predicate, finite-depth Bellman improvement, and a complete task-aware base-policy evaluation;
- 2) proofs of Markov sufficiency, preservation of a feasible task completion, base-policy no-degradation, and exact optimality on exhausted reduced graphs;
- 3) a directed SOOA representation with full-mesh visibility and motion audits, including complete GL Transmission Format (gTF) scene-node transforms, that admits safe connecting actions without requiring immediate inspection gain; and
- 4) a reproducible comparison on fixed required-target profiles, with matched baseline tasks, explicit coverage denominators, and a separate Robot Operating System 2 (ROS 2) execution evidence layer.

II. Related Work

Relative-motion models provide the dynamic foundation for orbital inspection. The Clohessy–Wiltshire equations remain a common baseline near a circular chief orbit because they are linear, interpretable, and inexpensive to propagate [8], [9]. Higher-fidelity flight design must account for nonlinear gravity, perturbations, navigation error, and actuator dynamics, but HCW propagation is useful for isolating how view ordering changes control effort and safety under a fixed model. Visibility-constrained orbit design has also been studied directly. Hou et al.

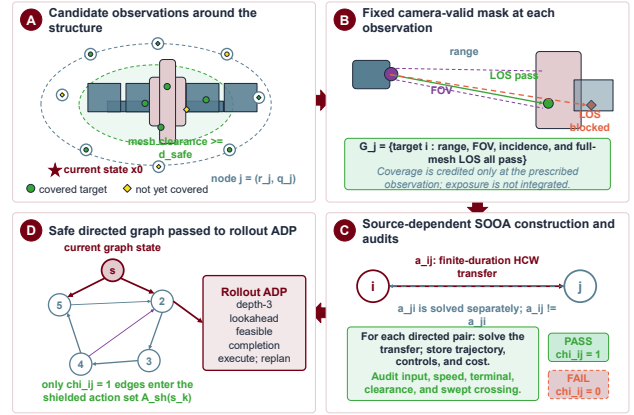


Fig. 1. OrbInspect formulation and data flow. (a) Candidate observations are generated around an International Space Station-scale structure outside the full-mesh safety margin. (b) Range, field of view, incidence, and full-mesh line of sight define the fixed visible-target mask G_j at each prescribed observation. (c) Each ordered node pair is assigned a separately solved finite-duration HCW transfer; its trajectory, controls, cost, sampled constraints, and swept-mesh crossing test determine the directed-edge indicator χ_{ij} . (d) Edges with $\chi_{ij} = 1$ form the safe directed SOOA graph whose shield-admissible actions are passed to the rollout-ADP policy in Fig. 2.

analyze relative orbits for visual inspection of the China Space Station [5], while Zhou et al. combine near-natural relative ellipses with formation control around a rotating station model [10]. These approaches provide structured orbital motion and stronger closed-loop control analysis than the present planner, but their primary object is an orbit or formation rather than a sequence of surface-coverage actions.

Robotic inspection provides the complementary view-selection perspective. Classical next-best-view work models sensor range, incidence, occlusion, reconstruction quality, and motion cost [4]. Sampling-based kinodynamic planners such as rapidly exploring random trees (RRTs) and asymptotically optimal variants incorporate differential constraints and collision avoidance in continuous state space [11], [12]. Spacecraft-specific extensions combine full-coverage inspection with kinodynamic planning for one or several inspectors [1], [2], and astrodynamics-informed sampling exploits natural relative motion during transfer search [6]. These methods are valuable when continuous path generation, coordination, or obstacle avoidance dominates. OrbInspect instead treats each audited transfer-observation pair as a finite decision action and carries the accumulated weighted coverage explicitly in the decision state.

Approximate dynamic programming and rollout address sequential decisions when exact Bellman recursion is too large [7], [13]. Rollout uses a base-policy completion cost to approximate the exact value and performs truncated Bellman policy improvement. Deep reinforcement learning (DRL) instead introduces learned function approximations for values or policies; a finite

predicted value alone does not certify that every required target can be reached. Constrained policy optimization controls expected constraint costs [14], whereas shielding removes actions that violate an independently specified safety model [15]. The present method combines an audited action shield with an explicit completion value. Its evaluated contribution is model-based rollout ADP; no trained deep policy or DRL performance claim is made for the required-target study.

Recent spacecraft-inspection studies increasingly couple geometry and motion. Apodaca et al. address inspection trajectory generation and coverage–energy trade-offs for in-orbit structures [3]. Zhou et al. study surface-aware tomographic paths around complex spacecraft geometry [16]. Ogri et al. address an execution-level problem involving image-feature tracking, information gain, and dwell under switched-system logic [17]. Together, these studies show that geometric coverage, orbital dynamics, and online sensing cannot be treated as fully independent. They also differ in the level at which the coupling occurs: orbit design, continuous path generation, online camera control, or discrete inspection planning.

OrbInspect is positioned at the discrete-action level. Its contribution is neither a new relative dynamics model nor a flight-certified collision-avoidance controller. Each SOOA is simultaneously a sensor-valid coverage contributor and a dynamically audited finite motion, so an infeasible transfer cannot improve the planner’s reported coverage. The algorithmic contribution is the coupling of rollout ADP with a finite-completion viability test on this inspection graph, not a claim that rollout itself is new. Formation-level studies provide complementary lessons: near-natural relative arcs can seed lower-effort SOOA libraries, and a rotating body frame should replace the fixed local-vertical/local-horizontal (LVLH) target assumption when station attitude motion is material. All optimality and safety statements remain conditional on the sampled library and stated audits.

III. System and Observation Model

The planner operates offline. It receives a known station mesh, a fixed nonempty required-target set, an initial relative state, camera parameters, a keep-out model, and actuation limits. It returns a time-indexed translational reference, a camera-boresight stream, selected SOOA records, accepted target identities, and summary metrics. Table I states the modeling assumptions and their implications for interpreting both evidence layers.

A. Relative Dynamics

The inspected structure is fixed in an LVLH frame attached to a nominal circular chief orbit. The axes follow the radial, along-track, and cross-track directions shown in Fig. 1. The chaser state is $\mathbf{x} = [\mathbf{r}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$, where $\mathbf{r} = [r_x, r_y, r_z]^\top$ and $\mathbf{v} = [v_x, v_y, v_z]^\top$. With mean motion n and commanded acceleration $\mathbf{u} = [a_x, a_y, a_z]^\top$, the HCW

model is

$$\begin{aligned} \dot{\mathbf{x}} &= A_{\text{HCW}}\mathbf{x} + B\mathbf{u}, \\ A_{\text{HCW}} &= \begin{bmatrix} 0 & I \\ K_n & C_n \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ I \end{bmatrix}, \\ K_n &= \text{diag}(3n^2, 0, -n^2), \quad C_n = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned} \quad (1)$$

For the continuous HCW model, piecewise-constant acceleration over Δt has the exact zero-order-hold map

$$\begin{aligned} \mathbf{x}_{k+1} &= A_d\mathbf{x}_k + B_d\mathbf{u}_k, \\ A_d &= e^{A_{\text{HCW}}\Delta t}, \\ B_d &= \int_0^{\Delta t} e^{A_{\text{HCW}}\tau} B d\tau. \end{aligned} \quad (2)$$

The reported implementation uses the same piecewise-constant input model but constructs and propagates each discrete boundary map with fixed-step fourth-order Runge–Kutta integration. We denote that numerical affine map by $\mathbf{x}_{k+1} = \tilde{A}_d\mathbf{x}_k + \tilde{B}_d\mathbf{u}_k$; (2) is the exact continuous-model reference, not a claim that the archived samples were generated by a matrix exponential. The transfer is therefore a finite-duration continuous-thrust maneuver with zero-order-hold acceleration, which may jump at sample boundaries, rather than an impulsive maneuver. Each segment is solved rest-to-rest at a candidate observation, and the numerical rollout is accepted only when its terminal position and velocity errors satisfy the stated tolerances.

B. Visibility, Prescribed Observation, and Coverage

The structure is represented by surface samples $\mathcal{T} = \{(p_i, n_i, a_i)\}_{i=1}^N$ with represented areas $a_i > 0$. Mission priority weights are separately defined as $w_i = a_i\eta_i$, where $\eta_i > 0$ is a priority multiplier; they are not geometric areas. A fixed nonempty set $\mathcal{K} \subseteq \mathcal{T}$ specifies the required inspection items before scenario evaluation. The camera attitude q_k defines unit boresight $b(q_k)$, and $\mathcal{O}$ denotes the occluding mesh. Target i is visible when

$$\begin{aligned} r_{\min}^{\text{cam}} &\leq \|p_i - \mathbf{r}_k\| \leq r_{\max}^{\text{cam}}, \\ \arccos \frac{(p_i - \mathbf{r}_k)^\top b(q_k)}{\|p_i - \mathbf{r}_k\|} &\leq \alpha_{\max}, \\ \arccos \frac{(\mathbf{r}_k - p_i)^\top n_i}{\|\mathbf{r}_k - p_i\| \|n_i\|} &\leq \theta_{\max}, \\ \text{LOS}(\mathbf{r}_k, p_i, \mathcal{O}) &= 1. \end{aligned} \quad (3)$$

The four tests exclude targets that are too near or too far, outside the field of view, observed at an excessive incidence angle, or blocked by station geometry. Line of sight is queried against a deterministic bounding-volume hierarchy containing every ISS mesh triangle; intersections at the target endpoint are excluded, while interior segment intersections block the view. In a closed-loop execution model, dwell could be accumulated as $\tau_{i,k+1} = \tau_{i,k} + \Delta t$ while target i remains visible and

TABLE I
Modeling assumptions and the boundary of the reported evidence

Assumption	Role in the baseline	Consequence for interpretation
Circular chief orbit	Supports linear HCW relative translation in the LVLH frame.	Quantitative results compare planners under the same local linear model, not a perturbed flight ephemeris.
Station fixed in LVLH	Keeps mesh targets and the mesh safety geometry stationary during one planning run.	Rotating appendages or station attitude motion require a body-frame extension.
Known mesh and state	Makes visibility, occlusion, and transfer evaluation deterministic.	Navigation, map, and shape uncertainty are not represented by the present safety margin.
Prescribed camera attitude	Points the boresight toward each candidate observation target and exports the corresponding quaternion stream.	Camera slew time, reaction-wheel limits, and closed-loop attitude error are outside the optimization.
Mesh-based sampled safety	Computes distance to the full International Space Station (ISS) triangle mesh at every rollout sample and rejects sampled motion segments that intersect the mesh.	Positive sampled clearance and surface nonintersection are planning audits rather than a robust continuous-time collision certificate.
No illumination or plume model	Isolates view geometry and translational feasibility.	Sun angle, shadow, contamination, and plume-impingement restrictions must be added for flight design.

accepted when $\tau_i \geq \tau_{\min}$. The offline study instead treats every candidate as a stabilized terminal observation with its prescribed exposure completed a priori. Selecting node j therefore credits its fixed visibility mask G_j , so $m_{k+1} = m_k \vee G_j$. The sampled-area coverage represented by mask m_k is

$$C^{\text{area}}(m_k) = \frac{\sum_i a_i m_{i,k}}{\sum_i a_i}. \quad (4)$$

Candidate generation can make a subset of the sampled surface invisible from every candidate observation. For the nonempty set $\mathcal{T}_{\text{insp}} = \cup_{j=1}^M G_j$, we also report priority-weighted inspectable coverage:

$$C_k^{\text{insp}} = \frac{\sum_{i \in \mathcal{T}_{\text{insp}}} w_i m_{i,k}}{\sum_{i \in \mathcal{T}_{\text{insp}}} w_i}. \quad (5)$$

Equation (4) estimates credited area over the original surface samples, whereas (5) is a priority-weighted score conditional on the candidate library. Their denominators remain fixed after scenario-specific observation loss. Neither is the primary required-target completion metric, which is

$$C^{\mathcal{K}}(m) = \frac{|\mathcal{K} \cap \text{supp}(m)|}{|\mathcal{K}|}, \quad (6)$$

$$\Gamma_{\mathcal{K}}(m) = \mathbf{1}\{\mathcal{K} \subseteq \text{supp}(m)\}.$$

Mission success requires $\Gamma_{\mathcal{K}} = 1$, even if optional area coverage is small. One observation can credit several required items, and several candidate viewpoints can satisfy the same item. The requirement is complete inspection of the prescribed samples under (3), not complete imaging of each physical assembly or defect detection. The reported mission duration includes transfer time but no additional dwell or exposure time. Attitude determines the visibility predicate and exported boresight; closed-loop attitude dynamics, illumination, image-resolution acceptance, and interruption of a dwell are outside the offline model.

C. Dynamic Feasibility and SOOAs

Every sampled state must satisfy

$$\|\mathbf{u}_k\|_2 \leq u_{\max}, \quad \|\mathbf{v}_k\|_2 \leq v_{\max}, \quad d_{\mathcal{S}}(\mathbf{r}_k) \geq 0, \quad (7)$$

where $d_{\mathcal{S}}(\mathbf{r}) = d_{\text{mesh}}(\mathbf{r}) - r_{\text{veh}} - d_{\text{safe}}$ is finite-body clearance above the required margin, d_{mesh} is unsigned center distance to the closest triangle in the complete ISS mesh, and $r_{\text{veh}} = 0.80 \text{ m}$ conservatively bounds the chaser visual model; thus $d_{\mathcal{S}} = 0$ is the sampled admissibility boundary. The mesh coordinates are obtained from the complete glTF scene hierarchy (a matrix or translation–rotation–scale at every ancestor), followed by the Simulation Description Format (SDF) pose and scale. In addition, each consecutive pair of rollout samples is tested for a triangle-surface intersection, and its continuous center-distance lower bound subtracts half the segment length from the smaller endpoint distance by the 1-Lipschitz property of distance to a closed set. This swept test catches a direct mesh crossing even if both endpoints have positive unsigned distance. For a directed rollout from source node i to destination node j , define $\Delta v_{ij} = \sum_k \|\mathbf{u}_{ij,k}\|_2 \Delta t$, root-mean-square distance e_{ij}^{rms} from the transfer samples to the destination position, and $d_{ij}^{\min} = \min_k d_{\mathcal{S}}(\mathbf{r}_{ij,k})$. The graph stage cost used in the reported study is

$$\ell_{ij} = \lambda_v \Delta v_{ij} + \lambda_e e_{ij}^{\text{rms}} + \lambda_s \max(0, -d_{ij}^{\min}) + \lambda_a. \quad (8)$$

For every shield-admissible edge, the clearance-violation term is zero. It remains in the shared cost implementation so deliberately unshielded comparison methods can be evaluated with the same expression.

A candidate observation node and a directed SOOA edge are, respectively,

$$v_j = (\bar{\mathbf{r}}_j, \bar{q}_j, G_j),$$

$$a_{ij} = (\mathbf{x}_{ij}[1:K], \mathbf{u}_{ij}[0:K-1], q_{ij}[1:K], \quad (9)$$

$$\Delta v_{ij}, d_{ij}^{\min}, u_{ij}^{\max}, v_{ij}^{\max}, e_{ij}^{\text{rms}}, e_{ij}^v, \rho_{ij}).$$

Here $\bar{\mathbf{r}}_j$ and $\bar{q}_j$ are the terminal observation pose and prescribed attitude, $G_j \subseteq \mathcal{T}$ is its camera-valid target

set, and e_{ij}^r and e_{ij}^v are terminal position and velocity errors. The stored state and attitude arrays contain the K post-step samples; the source state is held by node v_i . The initial condition is a distinguished source node v_0 with no visibility reward. For passive horizon $T_p = L_p \Delta t$, an optional audit can propagate zero-input HCW drift from every stored edge state and record $\rho_{ij} = \min_{1 \leq k \leq K, 0 \leq l \leq L_p} d_S(\mathbf{r}_{ij,k}^{\text{drift}}(l))$. The reported study disables this optional audit and therefore makes no passive-safety claim; it retains input, speed, full-mesh sampled clearance, swept surface-intersection, and terminal checks. Throughout this paper, “safe” means admissible under the audits explicitly enabled for the reported experiment. These deterministic checks do not certify robustness between samples or under model and state uncertainty.

IV. Viability-Preserving Rollout ADP

A. Candidate Graph and Markov State

Let $\mathcal{V} = \{v_0, v_1, \dots, v_M\}$ contain the initial node and the M candidate observation nodes in (9). A directed edge from v_i to v_j is the source-dependent SOOA a_{ij} , and it is absent when an enabled audit fails. Thus visibility G_j is node dependent, whereas trajectory, cost, clearance, input, terminal error, and passive margin are edge dependent. This separation is necessary because reaching the same observation from two different source nodes generally produces different HCW motion and audit values.

A node identifier is not a sufficient decision state because the value of a view changes after its targets have been covered. The finite-horizon Markov state is

$$s_k = (j_k, m_k, b_k, h_k), \quad (10)$$

where $j_k \in \{0, \dots, M\}$ is the current node, $m_k \in \{0, 1\}^N$ is the covered-target mask, $b_k \in \{0, 1\}^M$ is the selected-observation mask, and h_k is the remaining action budget. Selecting destination $a \in \{1, \dots, M\}$ traverses edge $a_{j_k a}$ and gives the deterministic transition

$$f(s_k, a) = (a, m_k \vee G_a, b_k \vee \mathbf{e}_a, h_k - 1). \quad (11)$$

The required set $\mathcal{K}$ is fixed problem data. The shield-admissible action set $\mathcal{U}_s(s)$ is defined below. The constrained graph problem minimizes total stage cost subject to (6), with zero terminal cost for a completed task and value $+\infty$ when no admissible completion exists within the remaining budget. Its Bellman recursion is

$$V^*(s) = \begin{cases} 0, & \Gamma_{\mathcal{K}}(m) = 1, \\ +\infty, & h = 0 \text{ or } \mathcal{U}_s(s) = \emptyset, \\ \min_{a \in \mathcal{U}_s(s)} Q^*(s, a), & \text{otherwise,} \end{cases} \quad (12)$$

$$Q^*(s, a) = \ell_{ja} + V^*(f(s, a)).$$

Here j and h are the current-node and remaining-budget components of s . Exact evaluation is exponential in the coverage and selected-node masks. The proposed method approximates this recursion with a short exact prefix and a computationally inexpensive, goal-aware terminal policy.

B. SOOA Construction and Safety Shield

Area-weighted targets are sampled from the mesh, and camera seeds are generated on offset shells around their outward normals. The binary visibility matrix $V_{ij} = \mathbf{1}\{i \in G_j\}$ is computed once from the terminal observation geometry. For a transition to $\bar{\mathbf{x}}_j = [\bar{\mathbf{r}}_j^T, \mathbf{0}^T]^T$, the planner solves the minimum-energy piecewise-constant HCW boundary-value problem under the numerical fourth-order Runge–Kutta (RK4) map

$$\begin{aligned} \min_{\mathbf{u}_0, \dots, \mathbf{u}_{K-1}} \quad & \sum_{k=0}^{K-1} \|\mathbf{u}_k\|_2^2 \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = \tilde{A}_d \mathbf{x}_k + \tilde{B}_d \mathbf{u}_k, \\ & \mathbf{x}_K = \bar{\mathbf{x}}_j. \end{aligned} \quad (13)$$

If the terminal influence matrix $\mathcal{B}_K = [\tilde{A}_d^{K-1} \tilde{B}_d \dots \tilde{B}_d]$ has full row rank, the stacked minimum-norm command from source state $\mathbf{x}_i^{\text{src}}$ is $\mathbf{U}_{ij}^* = \mathcal{B}_K^T (\mathcal{B}_K \mathcal{B}_K^T)^{-1} (\bar{\mathbf{x}}_j - \tilde{A}_d^K \mathbf{x}_i^{\text{src}})$. The implementation computes this expression through the 6×6 Gram system and then propagates the requested input without clipping. Let $\chi_{ij} = 1$ exactly when every stored edge sample satisfies (7), $e_{ij}^r \leq \varepsilon_r$, $e_{ij}^v \leq \varepsilon_v$, and $\rho_{ij} \geq 0$ when the passive audit is enabled. The shield then admits

$$\mathcal{U}_s(s) = \{a \in \{1, \dots, M\} : b_a = 0, \chi_{ja} = 1\}, \quad (14)$$

where j is the current-node component of s . Safe unvisited nodes remain admissible when they provide no immediate inspection gain, because a connecting action may be needed to reach a required target on the directed graph. The finite budget and no-revisit mask bound such connections. The shield makes an inadmissible edge unavailable to incumbent construction, local search, and exact recursion [15]. Its guarantee remains conditional on the sampled numerical rollout and deterministic model used to compute χ_{ij} .

C. Adaptive Base Policy and Rollout Value

Let $g_{\mathcal{K}}(s, a) = |\mathcal{K} \cap (G_a \setminus \text{supp}(m))|$ count newly accepted required targets, and let $\epsilon = 0.05$ prevent division by a near-zero stage cost. For the required-target task, the deterministic adaptive base policy is

$$\mu(s) \in \arg \max_{a \in \mathcal{U}_s(s)} \left(\frac{g_{\mathcal{K}}(s, a)}{\max\{\epsilon, \ell_{ja}\}}, g_{\mathcal{K}}(s, a), -\ell_{ja} \right), \quad (15)$$

where the tuple is compared lexicographically, followed by deterministic node-identifier tie breaking. The policy recomputes gain after every accepted observation; zero-gain ties favor lower stage cost. Starting from state s , its completion value is

$$\hat{V}_0(s) = \begin{cases} \sum_{k=0}^{L-1} \ell_{j_k \mu(s_k)}, & \mu \text{ reaches } \Gamma_{\mathcal{K}} = 1 \text{ within } h, \\ +\infty, & \text{otherwise.} \end{cases} \quad (16)$$

Thus $\hat{V}_0$ is not a learned critic. It is a model-based approximate cost-to-go obtained by simulating a complete, shielded base policy.

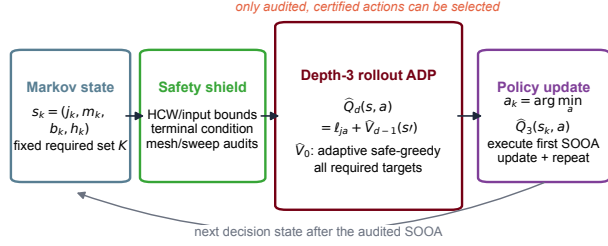


Fig. 2. Proposed viability-preserving rollout ADP. The finite graph state carries current node, covered and selected masks, and remaining budget, with required set $\mathcal{K}$ fixed as problem data. The shield removes inadmissible SOOAs. A depth- d exact prefix is terminated by a complete task-aware base-policy simulation; a finite branch value certifies $\Gamma_{\mathcal{K}} = 1$ within the budget. The first minimizing action is executed and the value is recomputed.

For rollout depth $d \geq 1$, define the truncated Bellman value

$$\hat{V}_d(s) = \begin{cases} 0, & \Gamma_{\mathcal{K}}(m) = 1, \\ +\infty, & h = 0, \\ \min_{a \in \mathcal{U}_s(s)} \hat{Q}_d(s, a), & \text{otherwise,} \end{cases} \quad (17)$$

where $\hat{Q}_d(s, a) = \ell_{ja} + \hat{V}_{d-1}(f(s, a))$. The minimum over an empty action set is $+\infty$. Only actions with finite completion value participate in the minimization. The rollout policy selects the minimizing first action, applies (11), and rebuilds the depth- d tree at the new state. This repeated policy-improvement step distinguishes the proposed method from following a fixed incumbent suffix. State-depth memoization removes duplicate subproblems within one decision. The following rollout guarantees require exhaustive evaluation of this prefix and the same deterministic Markov base policy at every leaf; they do not apply to a pruned learned-policy search. Fig. 2 shows how the finite-completion test, safety shield, and repeated depth- d improvement interact.

D. Properties and Exact Oracle

PROPOSITION 1 (Monotone submodularity of fixed-mask coverage) *For an observation set $S \subseteq \{1, \dots, M\}$, let $F(S) = \sum_{i=1}^N w_i \mathbf{1}\{i \in \cup_{j \in S} G_j\}$. If all $w_i \geq 0$, then F is normalized, monotone, and submodular.*

Proof:

Clearly $F(\emptyset) = 0$. If $A \subseteq B$, every target covered by A is covered by B , so nonnegative weights give $F(A) \leq F(B)$. For $a \notin B$, the marginal gain $F(A \cup \{a\}) - F(A)$ is the total weight in $G_a \setminus \cup_{j \in A} G_j$. Since $A \subseteq B$, this set contains $G_a \setminus \cup_{j \in B} G_j$, proving diminishing returns. ■

Remark 1 (Role of submodularity) Proposition 1 also holds for required-target gain by assigning unit weight to $\mathcal{K}$ and zero elsewhere. It does not justify removing zero-gain connecting actions or supply a greedy approximation ratio for a directed inspection tour. Both SOOA cost and

Algorithm 1 Viability-preserving rollout ADP

Require: Candidate graph, shield $\mathcal{U}_s$, required set $\mathcal{K}$, budget H , depth d

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1:  $s \leftarrow (0, \mathbf{0}, \mathbf{0}, H)$ ;  $\pi \leftarrow ()$ 
2: while  $\Gamma_{\mathcal{K}}(m) = 0$  and  $h > 0$  do
3:   Enumerate every  $a \in \mathcal{U}_s(s)$ 
4:   for each admissible  $a$  do
5:      $\hat{Q}_d(s, a) \leftarrow \ell_{ja} + \hat{V}_{d-1}(f(s, a))$ 
6:   end for
7:   Discard every action with  $\hat{Q}_d(s, a) = +\infty$ 
8:   if no finite action remains then
9:     return no certified completion
10:  end if
11:   $a^* \leftarrow \arg \min_a \hat{Q}_d(s, a)$  with deterministic ties
12:  Append  $a^*$  and its audit record to  $\pi$ 
13:   $s \leftarrow f(s, a^*)$ 
14: end while
15: return  $\pi$  if  $\Gamma_{\mathcal{K}}(m) = 1$ ; otherwise return no certified completion

```

the admissibility of the next observation depend on the current node.

PROPOSITION 2 (Markov sufficiency) *For a fixed deterministic candidate graph with fixed node masks, required set $\mathcal{K}$, source-dependent edge records, target weights, revisit rule, and horizon, the state in (10) is Markov for the finite graph decision problem.*

Proof:

The current node j_k identifies the source of every possible next edge; m_k and fixed $\mathcal{K}$ determine the remaining inspection requirements; b_k determines which observations remain selectable under the no-revisit rule; and h_k determines the remaining budget. By assumption, visibility masks and edge costs and audits are deterministic records indexed only by the current and destination nodes. Equations (11), (12), and (14) therefore depend on no earlier history once (j_k, m_k, b_k, h_k) is given. The proposition concerns binary prescribed-observation acceptance; partial dwell, multiple independent look directions, or time-dependent illumination would require additional state. ■

PROPOSITION 3 (Compositional preservation of stored audits) *Consider a sequence $(a_{j_{k-1}j_k})_{k=1}^L$ whose first source is v_0 , whose idealized rest-to-rest endpoints match exactly, and whose every edge has $\chi_{j_{k-1}j_k} = 1$. Then every stored sample of the concatenated sequence satisfies the input, speed, and clearance inequalities, every edge endpoint satisfies its terminal conditions, and every stored passive-drift rollout has nonnegative margin when that audit is enabled.*

Proof:

By the definition of χ_{ij} , all stated conditions hold on each accepted edge record. Exact endpoint matching identifies the terminal state of edge k with the source state of edge $k+1$, so the records concatenate without an unmodeled

jump. The sample set of the complete trajectory is the finite union of the per-edge sample sets; therefore every member of that union retains the corresponding per-edge audit result. ■

Remark 2 (Meaning of audit preservation) The proposition is a finite-record composition result, not a continuous-time barrier certificate. It does not cover inter-sample clearance, endpoint perturbations beyond tolerance, closed-loop tracking error, or robustness to model and state uncertainty.

PROPOSITION 4 (Rollout viability) *For the deterministic finite graph, if $\hat{V}_d(s) < +\infty$, Algorithm 1 selects a shield-admissible action whose successor has a finite goal-reaching completion within the remaining budget. Replanning preserves this property until the goal is reached.*

Proof:

Finiteness in (17) provides a certificate consisting of shield-admissible prefix actions followed, unless the goal is reached earlier, by a base-policy completion. The selected first action is part of such a certificate, so its successor retains a goal-reaching sequence within the reduced budget. When the depth- d tree is rebuilt, the certificate tail remains admissible; if another exact prefix action is required, the next action of the certified base completion supplies it. Determinism preserves the recorded masks, costs, and audits. Induction over the decreasing action budget therefore proves goal attainment. ■

PROPOSITION 5 (Base-policy no-degradation) *If the adaptive base policy reaches the goal from s , then $\hat{V}_d(s)$ is finite and $\hat{V}_d(s) \leq \hat{V}_0(s)$ for every $d \geq 1$. For a fixed depth d , the realized graph cost of repeated replanning in Algorithm 1 is no greater than $\hat{V}_d(s)$ and therefore no greater than the base-policy cost.*

Proof:

The first d base-policy actions, truncated if the goal is reached earlier, followed by the remaining base completion form a feasible certificate in (17); hence $\hat{V}_d(s) \leq \hat{V}_0(s)$. Now take a minimizing depth- d certificate at a replanning state and let s' be the successor of its first action a . The certificate tail, extended when necessary by the next base-policy action, is feasible in the rebuilt depth- d tree at s' . Therefore $\ell_{ja} + \hat{V}_d(s') \leq \hat{V}_d(s)$. Summing this inequality over successive replans telescopes to a realized cost no greater than the initial $\hat{V}_d(s)$. The result concerns the graph stage cost; reductions in physical Δv are evaluated separately. ■

For reduced-graph certification, exact recursion caches (12) for every reachable tuple (j, m, b, h) and enumerates every action in $\mathcal{U}_s(s)$ rather than a local reversal neighborhood.

THEOREM 1 (Optimality on an exhausted reduced SOOA graph) *For fixed finite nodes, deterministic visibility masks, deterministic audited edge costs, a required set $\mathcal{K}$, the no-revisit rule, and horizon H , exhaustive evaluation*

of (12) returns a minimum-cost task-completing shield-feasible sequence when one exists. If no such sequence exists, it returns value $+\infty$ and reports infeasibility.

Proof:

Proceed by induction on the remaining budget h . At $h = 0$, (12) returns zero exactly for a goal state and $+\infty$ otherwise, which is correct. Assume the recursion is correct for all states with at most $h - 1$ remaining actions. At a state with h actions, every feasible sequence begins with exactly one $a \in \mathcal{U}_s(s)$ and then a feasible continuation from $f(s, a)$; by the induction hypothesis, the recursion assigns that continuation its minimum cost or $+\infty$ if none exists. Minimizing over every admissible first action therefore returns the minimum total cost, or $+\infty$ when all continuations are infeasible. The claim is limited to the exhausted sampled graph and does not extend to omitted camera poses or continuous-space trajectories. ■

Remark 3 (Graph certificate versus mission optimality) Theorem 1 certifies only the nodes, edges, masks, and audits included in the exhausted graph. Omitted camera poses, alternative transfer durations, and continuous-space trajectories remain outside the certified decision set.

With maximum branching factor B , depth d , and at most H base-policy actions, a direct decision evaluation is $O(HB^{d+1})$ in the worst case; memoization is effective when different prefixes reach the same mask state. The reported $d = 3$ setting is an offline planner. Table III reports a post-selection, validation-only diagnostic over $d = 1, \dots, 6$ for the fixed required task. Its platform-specific timing is descriptive and does not establish real-time capability.

V. Experimental Protocol

A. Fixed Geometry and Required-Target Profiles

The study uses the corrected public ISS mesh distributed by the National Aeronautics and Space Administration (NASA) [18]. Its complete transformed geometry contains 247 525 triangles. Ninety equal-represented-area samples define the original surface denominator; 41 are observable from at least one of the 24 retained candidate observations. The directed archive contains 24 initial-to-observation and 24×23 inter-observation records, of which 264 pass every enabled gate. Every method receives the same graph and scenario-specific available-node set.

Required-target profiles are selected deterministically from target geometry before evaluating the new scenarios. Their IDs, coordinates, selection rule, visibility masks, and checksums are stored in the mission manifest. These are synthetic, spatially distributed inspection requirements within the fixed observable library. The mesh sampler does not retain verified component-criticality annotations, so the profiles are not represented as an operational ISS maintenance checklist. Full completion means acceptance of all prescribed samples under (3); it does not establish

full-region imagery, physical defect detection, or complete ISS surface inspection.

The required set is unchanged when a scenario removes observation nodes. If a required item disappears from the union of all available visibility masks, the task is ruled out by visibility. Passing this necessary condition does not prove the existence of a directed safe route within the budget. Feasibility is known when a method constructs a valid route, or when an exhausted exact search provides a certificate; otherwise the outcome remains unresolved rather than being declared structurally feasible.

B. Dynamics, Costs, and Observation Accounting

Each rest-to-rest SOOA lasts 90 s and uses fixed-step fourth-order Runge–Kutta (RK4) propagation with $\Delta t = 3$ s. The HCW mean motion is 1.13137×10^{-3} rad/s, the acceleration limit is $u_{\max} = 0.060$ m/s², and the speed limit is $v_{\max} = 2.0$ m/s. The vehicle radius is 0.80 m, the full-mesh safety margin is 2 m, and terminal position and velocity tolerances are 0.5 m and 0.05 m/s. Equation (8) retains $(\lambda_v, \lambda_e, \lambda_s, \lambda_a) = (5, 0.18, 120, 0.05)$. The action budget is 14. Required-only completion has no additional global area threshold.

Coverage is computed from unions of target identities, with duplicate credit excluded. We report required-target completion $C^{\mathcal{K}}$, priority-weighted inspectable score C^{insp} , and the unweighted fractions over 41 observable and 90 original samples. Because the original sample areas are equal, the last fraction is the archived sampled-area estimate; scenario priority multipliers do not change that area denominator. Prescribed terminal exposures are assumed complete in the offline model, and no extra dwell, image-resolution gate, or illumination process is simulated.

C. Development History and Fresh Confirmation

The earlier 80% survey and its post-selection diagnostics are development history; their cost improvements are not transferred to this mission. A first required-target campaign froze the three geometry-defined profiles and evaluated 36 validation, 150 test, and 90 shifted scenarios without feasibility rejection. The nine-item profile was designated primary before evaluation. On that profile, depth-three ADP completed 37/50 test and 10/30 shifted tasks, whereas greedy construction and its fixed-suffix and local-search descendants completed none. Those failed initializations cannot support a matched-completion cost comparison. Table IV retains the first-campaign outcomes rather than replacing them.

A subsequent fresh confirmation therefore compares depth-three ADP against one-step ADP and local search initialized by a completed one-step ADP route. Its methods, penalty, scenario inventory, and implementation hashes are frozen before evaluating any of its new outcomes. The nine prescribed targets, depth-three policy, dynamics, and cost remain unchanged. The confirmation

contains 12 validation, 50 test, and 30 shifted scenarios, with seed identities disjoint from the first campaign and archived development runs. No target is removed or replaced after seeing outcomes. The first campaign is explicitly development evidence for the comparator choice, not an untouched confirmation set.

Each scenario samples a node-removal probability uniformly from $[0, 0.10]$ and independently retains or removes each candidate using that probability. It multiplies the priorities of approximately 6%–14% of targets by a factor in $[2, 4]$. The shifted distribution uses removal probabilities in $[0.15, 0.25]$, priority fractions of 12%–20%, and factors in $[5, 8]$. Every sampled scenario is retained, including loss of all views of a required target. Priorities affect background scores and the coverage-only ablation, but not the unweighted required-target goal or stage cost. These are perturbations of one frozen graph, not independent geometries; distinct seeds can yield identical routes.

The frozen depth-three setting uses the same deterministic task-aware base policy as the incumbent. After the confirmation analysis, the validation diagnostic was extended to depths one through six to assess the cost–workload trade-off beyond the chosen depth. All six depths were rerun sequentially on the same 12 validation scenarios, fixed nine-target requirement, graph, action budget, and implementation. Each depth retains every scenario, including missing required visibility. This extension evaluated no additional test or shifted scenarios; depths four through six remain validation-only, and the frozen primary comparison is unchanged. Every depth uses the same unvisited, shield-admissible action set, including zero-gain connecting observations.

Timing measures complete-route generation, including the preliminary base-policy call, in Python 3.14 on an Apple M4 host running macOS; it excludes graph loading and result serialization. There is one timing observation per scenario–depth pair in an ascending-depth sequential pass. These measurements characterize the present implementation and include uncontrolled scheduling and thermal effects; they are neither repeated-timing uncertainty estimates nor onboard deadlines. Safety-screen counts exclude the separate preliminary base-policy call.

All six depths complete the same nine of 12 validation tasks; the other three lack visibility of at least one required item. On this common nine-task cohort, increasing depth from two to three reduces mean graph cost by 11.19%, from 133.450 to 118.520. Depth three captures 84.11% of the observed reduction from depth one to the lowest mean cost among depths one through six. Depth four further reduces mean cost by 2.74%, to 115.272, but raises median planning time from 1.009 s to 4.676 s and mean safety-screen count by a factor of 4.74. Depths five and six yield no further cost or completion improvement, while their median times increase to 17.660 s and 53.013 s, respectively. These are 17.50 and 52.52 times the depth-three median; the corresponding screen-count ratios are 17.86 and 54.15.

TABLE II

Frozen synthetic required-target profiles. Three equal-width slabs span the mesh longest axis; each slab uses centroid-first, normalized farthest-point selection. IDs omit the common mesh prefix. The nine-item profile is primary

Profile	Items	Slab counts	Required sample IDs
required06	6	2/2/2	00055, 00052, 00007, 00001, 00033, 00015
required09	9	3/3/3	00055, 00052, 00059, 00007, 00001, 00079, 00033, 00015, 00019
required12	12	4/4/4	00055, 00052, 00059, 00064, 00007, 00001, 00079, 00045, 00033, 00015, 00019, 00024

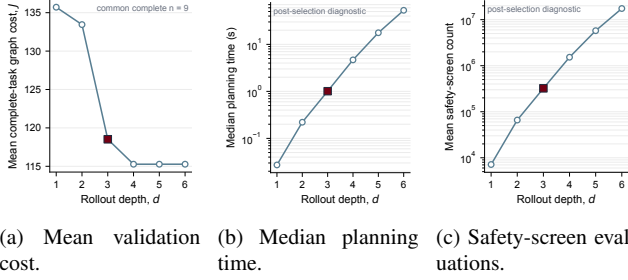


Fig. 3. Post-selection required-target validation diagnostic for depths one through six. The square marks the unchanged depth-three primary setting. Cost uses the same completed-task cohort at all depths; runtime and safety-screen workload retain all 12 cases. Panels (b) and (c) use logarithmic scales. The accompanying table reports completion counts and changes relative to depth three. This diagnostic quantifies a practical cost-workload trade-off, not a universally optimal depth or real-time capability.

These observations support retaining the frozen depth-three policy as a computational compromise, not as the minimum-cost or uniquely optimal depth. If the additional computation is acceptable, depth four provides lower cost in this diagnostic. The plateau at depths four through six is empirical, not an exact-optimality certificate, and realized receding-policy costs need not decrease monotonically with depth.

D. Matched Baselines and Statistical Reporting

The confirmation includes the task-aware greedy incumbent, one-step ADP, ADP-seeded local search, and depth-three ADP. All use the identical required-target predicate, dynamics, safety gates, and budget. One-step ADP uses (17) with $d = 1$. The local comparator applies three audited best-improvement reversal passes to that completed route and retains its feasible seed if improvement fails. It is an ADP-initialized sequence baseline, not an independent non-ADP planner or an exhaustive solver. The proposed $d = 3$ policy uses the original greedy base value and receives no fallback from the one-step route. This comparison separates repeated Bellman improvement from improving a feasible ordering.

An additional coverage-only policy retains the earlier 80% priority-weighted stopping rule but is evaluated against the same fixed required-target identities. It is an objective ablation, not a matched complete-task cost baseline. This comparison tests whether satisfying an aggregate score also satisfies the explicit task. The earlier

fitted-critic results belong to the earlier objective and are not reported as a trained DRL comparison for this campaign.

Primary reporting distinguishes all-required mission success from cost conditional on completion. Success and missing-target counts use every generated scenario. Graph cost and physical Δv are compared within jointly successful depth-three ADP/seeded-local pairs. The all-scenario diagnostic is $J_{\text{pen}} = J + 500(1 - C^K) + 500(1 - \Gamma_K)$; its fixed penalties are evaluation choices, not physical maneuver costs. Percentile intervals for paired mean differences use 10 000 resamples; exact two-sided sign tests exclude numerical ties. Intervals characterize this scenario generator, not flight or external-mesh generalization. On-line time includes route initialization, including the one-step seed for local search. Safety-screen counts describe the selected planner call and exclude the separately timed initializer.

VI. Results

A. Complete Required-Target Inspection

In the fresh nine-target confirmation, depth-three ADP completed every prescribed target in 43/50 test scenarios and 9/30 shifted scenarios. Every successful route achieved $C^K = 100\%$, irrespective of optional background coverage. These are counts of complete missions, not a claim that all generated tasks were successful. The corresponding seeded-local counts were 43/50 and 8/30. The separate first-campaign profile rates use different scenario seeds and are descriptive, not paired estimates of increasing requirement difficulty.

In the confirmation, all 43 test and nine shifted scenarios retaining visibility of every required item have an actual depth-three ADP completion witness. Every remaining scenario loses all available views of at least one mandatory item and is therefore impossible in the fixed library. This agreement is an observed result, not a general sufficiency theorem for the union-of-views test. In the first campaign, one six-target shifted case retains the required visibility but yields no ADP certificate; its directed feasibility remains unresolved. Proposition 4 applies to finite completion certificates and does not guarantee finding one for every physically feasible task.

TABLE III

Post-selection required-target depth diagnostic on the same 12 validation scenarios. Graph-cost means use the 9 scenarios completed at all six depths. Time and screen counts include all 12 cases; ratios use depth three. Negative ΔJ denotes lower mean cost. All depths were rerun sequentially; the primary depth-three test results are unchanged

Depth	Complete	Mean J	ΔJ (%)	Median time (s)	Time ($\times$)	Mean screens	Screens ($\times$)
1	9/12	135.714	+14.51	0.0272	0.03	7,132	0.02
2	9/12	133.450	+12.60	0.2207	0.22	65,928	0.20
3	9/12	118.520	+0.00	1.0094	1.00	322,349	1.00
4	9/12	115.272	-2.74	4.6760	4.63	1,527,408	4.74
5	9/12	115.272	-2.74	17.6601	17.50	5,757,081	17.86
6	9/12	115.272	-2.74	53.0133	52.52	17,454,611	54.15

TABLE IV

First-campaign development results by fixed profile. Local search in this campaign was initialized by the greedy incumbent, unlike the fresh confirmation comparator. Profiles use different seeds, so cross-profile rates are descriptive

Profile	Required items	Split	ADP complete	Local complete
required06	6	Test	40/50	0/50
required09	9	Test	37/50	0/50
required12	12	Test	38/50	0/50
required06	6	Shifted	10/30	0/30
required09	9	Shifted	10/30	0/30
required12	12	Shifted	10/30	1/30

TABLE V

Required-target results. Resource means use successful routes of each method; the final column includes every scenario with the fixed failure penalty

Split	Method	Complete	Mean SOOAs	Mean Δv (m/s)	Mean J	Penalized J
Test	Task-aware incumbent	0/50	—	—	—	911.242
Test	One-step ADP	43/50	8.21	16.512	135.945	256.913
Test	ADP-seeded local search	43/50	8.21	15.777	129.892	251.707
Test	Depth-three ADP	43/50	9.00	14.033	115.641	239.451
Shifted	Task-aware incumbent	0/30	—	—	—	851.059
Shifted	One-step ADP	8/30	9.25	17.012	140.102	770.694
Shifted	ADP-seeded local search	8/30	8.88	15.888	130.841	768.224
Shifted	Depth-three ADP	9/30	9.11	14.656	120.751	736.225

B. ADP Policy Improvement at Matched Completion

The predeclared all-scenario endpoint is the ADP-minus-seeded-local penalized score. Its mean differences are -12.256 on the 50 test scenarios and -31.999 on the 30 shifted scenarios, with 95% intervals $[-14.155, -10.212]$ and $[-91.494, -1.241]$, respectively. These results use the stated failure penalties and do not imply physical-resource savings on failed routes. The following complete-task comparisons provide the separate resource interpretation.

There were 43 jointly successful test pairs and 8 jointly successful shifted pairs. The mean ADP-minus-local graph-cost differences were -14.251 and -9.890 , respectively. The corresponding physical Δv differences were -1.744 m/s and -1.207 m/s. Table VI reports the paired uncertainty and wins/ties/losses. The cost comparison is conditional on both methods completing the task and must be read with the full-population success counts.

On the 43 complete test pairs, mean Δv decreases from 15.777 m/s to 14.033 m/s, an 11.05% reduction in the ratio of means. The paired reduction is 1.744 m/s, with a 95% interval of 1.525–1.923 m/s. ADP wins 40 pairs and loses three; the guarantee does not require winning against local search. The eight shifted pairs are a smaller descriptive sample: their mean reduction interval excludes zero, but the exact two-sided sign test gives $p = 0.0703$. These endpoints and tests are not interchangeable claims of significance.

The theoretical comparator is the deterministic adaptive base policy. Proposition 5 bounds graph cost relative to that policy when it completes; it does not imply superiority over every local-search route or a lower physical Δv in every scenario. The greedy base completes no confirmation task, so an empirical base-cost-bound check on this campaign would be vacuous. Finite-graph unit tests separately exercise the bound, useful connecting actions, and exact recursion; no new mesh-based exact-oracle campaign is claimed. The present comparison in-

TABLE VI

ADP-minus-seeded-local effects. The predeclared penalized endpoint includes all scenarios; physical resources and graph cost use jointly complete pairs. Intervals are percentile 95% paired-bootstrap intervals; sign tests exclude ties

Split	Endpoint	Pairs	Mean difference	95% interval	Win/tie/loss	Sign-test p
Test	J_{pen} (all)	50	-12.256	$[-14.155, -10.212]$	40/7/3	3.02×10^{-9}
Test	J	43	-14.251	$[-15.715, -12.461]$	40/0/3	3.02×10^{-9}
Test	Δv (m/s)	43	-1.744	$[-1.923, -1.525]$	40/0/3	3.02×10^{-9}
Shifted	J_{pen} (all)	30	-31.999	$[-91.494, -1.241]$	8/21/1	0.0391
Shifted	J	8	-9.890	$[-14.111, -4.688]$	7/0/1	0.0703
Shifted	Δv (m/s)	8	-1.207	$[-1.713, -0.575]$	7/0/1	0.0703

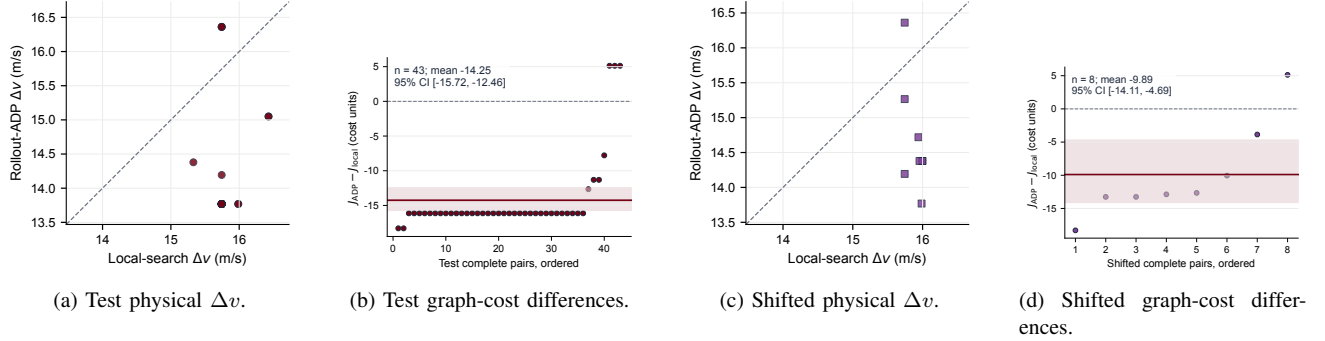


Fig. 4. Required-target ADP/local comparison on jointly successful pairs: 43 test and 8 shifted scenarios. Dashed lines denote parity. Bands and solid horizontal lines in (b) and (d) denote 95% paired-bootstrap intervals and mean graph-cost differences. Failed missions are excluded from these physical-cost panels and retained in the success tables and all-scenario penalty analysis.

TABLE VII

Required-target completion under the coverage-only objective. The required set is identical to that of the main comparison

Split	Objective ablation	All required complete
Test	coverage only80	1/50
Shifted	coverage only80	0/30

stead measures empirical improvement over completion-capable one-step and seeded-local policies.

C. Objective Ablation and Safety

The coverage-only ablation is evaluated using the new required-target denominator even though its stopping rule remains a global score. A route that reaches 80% background score while omitting a required item fails the prescribed inspection task. Conversely, a required-only route can finish with a lower background score when every required item has been accepted. The ablation quantifies this distinction without redefining targets from the resulting routes.

The coverage-only policy completes the nine-item requirement in only 1/50 test and 0/30 shifted scenarios. In contrast, 41 of the 43 completed ADP test missions finish below 80% weighted background coverage while accepting all nine prescribed items; their mean background score over all 43 successes is 72.32%. Thus the lower background score is consistent with complete

execution of the declared sample-level task, not evidence that the remaining mesh has been inspected.

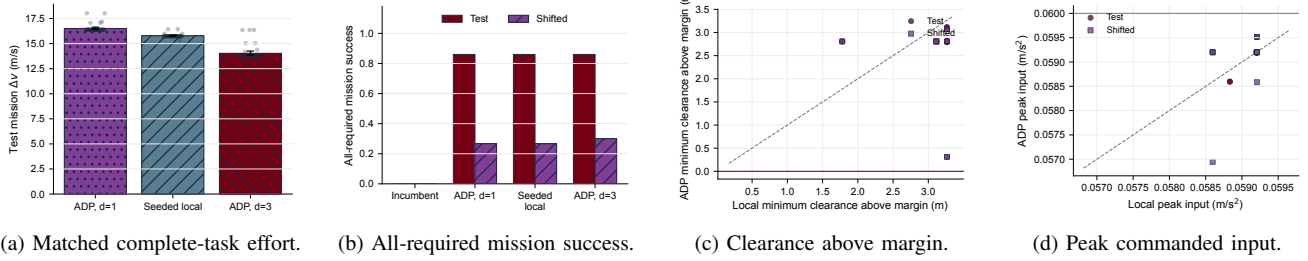
The observed safety quantities are properties of the accepted deterministic edge records. Full-mesh occlusion and collision checks still include structure outside the required target set. Restricting the inspection task therefore does not reduce the safety geometry. Timing and evaluation counts remain descriptive measurements of the present implementation, not onboard real-time benchmarks.

D. Representative Required-Target Route

The illustrative pair is scenario `required09_test_000`, selected as the jointly complete test pair nearest the median ADP-minus-local graph-cost difference (deterministic ID tie break). Both routes complete all 9 prescribed targets. Rollout ADP uses 9 SOOAs, $J = 113.472$, and $\Delta v = 13.769$ m/s; local search uses 8 SOOAs, $J = 129.605$, and $\Delta v = 15.744$ m/s. The respective background inspectable scores are 77.22% and 64.97%. Repropagation reproduces both archived costs and velocity increments to numerical tolerance.

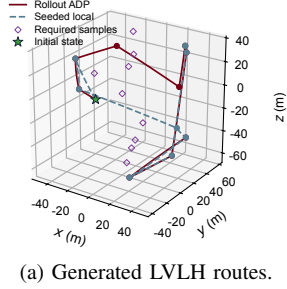
E. Execution Interface and Historical ROS Evidence

The target-credit implementation retains per-observation target IDs and forms the union of accepted observations. Planned cumulative coverage is not

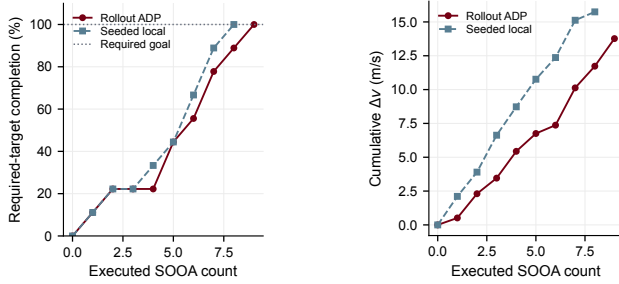


(a) Matched complete-task effort. (b) All-required mission success. (c) Clearance above margin. (d) Peak commanded input.

Fig. 5. Fresh nine-target component and safety audit. Panel (a) uses the 43 test scenarios completed by one-step ADP, seeded local search, and depth-three ADP, with mean and 95% bootstrap interval. The greedy incumbent is reported in the all-scenario success panel (b). Panels (c) and (d) use jointly complete depth-three ADP/seeded-local pairs. No passive-drift or uncertainty-robustness claim is made.



(a) Generated LVLH routes.



(b) Required-target completion. (c) Cumulative physical Δv .

Fig. 6. Illustrative required-target case selected by a stated median-effect rule from jointly successful test pairs. Both methods complete the same fixed required set; background coverage is a separate descriptor. Markers identify terminal observations and required samples, and the star is the common initial state. The case illustrates the policy mechanism and does not replace the all-scenario comparison.

imported when a later observation passes. Required completion, duplicate credit, rejected observations, and export consistency are tested independently of ROS. The new schema carries the fixed requirement manifest and separate planned and accepted metrics while preserving existing message interfaces.

An offline replay export of the illustrative confirmation ADP route preserves its nine actions, all nine required IDs, and 77.22% planned weighted background coverage. Target unions and resource metrics match the archived route. The six core CSV streams distinguish planned trajectories, controls, coverage, safety, planning decisions, and mission events from executed measurements. This materialization checks the new mission interface but is not evidence that those observations were executed by ROS.

The available Ubuntu 24.04.4/ROS 2 Jazzy demonstration predates the required-target campaign and uses the earlier 80% survey objective. It is retained solely as an execution/interface example for the corrected mesh. No new required-target ROS campaign was executed in the present revision, so the new primary results are offline ADP results. A subsequent required-target execution must use the new manifest, credit each prescribed item, and repeat the finite-body audit before supporting an execution-level completion claim.

F. Historical Corrected ROS 2 Survey Execution

The historical 80%-survey validation task ran on Ubuntu 24.04.4 with ROS 2 Jazzy, Gazebo Harmonic, and RViz2. Gazebo supplied the station/chaser rendering and camera image only; the ROS-native HCW node remained the state source of truth. The input route was reconstructed from the corrected 24-node graph using the complete glTF scene hierarchy and a conservative $r_{veh} = 0.80$ m chaser bounding radius. The route manifest records 10 planned SOOAs, 900 s mission duration, 81.33% planned coverage, and 2.808 m minimum planned finite-body clearance above the required 2 m margin.

The $5\times$ graphical closed-loop execution credited all 10 terminal observations, reached 81.33% coverage, and completed at mission time 900.01 s without a failed action. The post-run audit evaluated 18918 recorded trajectory samples and every intervening segment against all 247525 fully transformed ISS triangles. No segment crossed the mesh. The smallest sampled center distance was 8.506 m; after the continuous segment lower bound and the 0.80 m vehicle radius, the minimum body distance was 7.692 m, or 5.692 m above the required safety margin. Peak filtered acceleration was 0.0600 m/s². Terminal, reference-stream, data-presence, acceleration, full-mesh, swept-crossing, and finite-body gates all passed.

The retained evidence directory contains the six core CSV streams, configuration and route manifests, launch log, summary, version-2 mesh audit, raw RViz and camera streams, an annotated H.264 video, and a milestone preview. The 198.3-s video uses RViz2 as the main view and the live `/chaser/camera/image` topic as an inset. It is an execution and interface demonstration, not an

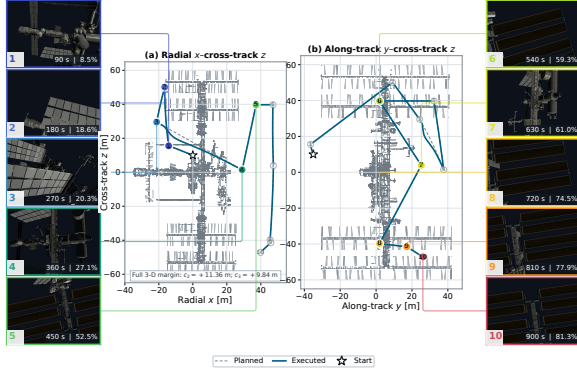


Fig. 7. Historical survey execution: trajectory-linked camera views from the accepted corrected `validation_002` task with an 80% inspectable-coverage goal. Views 1–5 link locally to the radial-cross-track (x – z) projection, and views 6–10 link to the along-track-cross-track (y – z) projection; hollow gray markers retain the other observations as context. Inset labels give mission time and cumulative coverage. Dashed gray and solid blue denote planned and executed paths, and the star is the initial state. Apparent overlap in either projection does not imply contact: the annotated clearances and collision decision use the full 3-D transformed mesh and swept path. Gazebo rendered the views, while ROS HCW dynamics propagated state; coverage came from frozen geometric visibility masks, not image pixels. This task predates the required-target study and does not validate its new mission profiles.

additional offline paired-comparison observation. Fig. 7 links all ten synchronized camera frames to their exact credited positions in two orthogonal projections of the recorded closed-loop trajectory.

The earlier paired ROS campaign used a translation-only interpretation of the gITF node hierarchy. Re-auditing its illustrative route with the complete hierarchy exposed a swept intersection with the S4 truss and a negative finite-body clearance. Those numerical ROS campaign claims and their old video are therefore excluded rather than carried into the corrected manuscript. The current result establishes consistency of one complete closed-loop task with the corrected geometry and audit; paired ROS method superiority would require a new full campaign from the corrected route bundle.

VII. Discussion

The central contribution is the ADP policy-improvement mechanism under an explicit inspection-completion constraint. The approximate terminal value is the complete cost of a tractable base policy. A finite Bellman prefix searches beyond that policy’s immediate choice while retaining a checkable goal-reaching completion. This construction gives the viability and base-cost statements their meaning; labeling an arbitrary scalar coverage score as complete inspection would not provide the same task guarantee.

The required-target formulation changes the application criterion without requiring a new coverage state for every fixed mission. Covered identities and the frozen required set determine the residual task. The same theoret-

ical construction applies to a hybrid goal that additionally imposes a background threshold, provided the base policy, rollout boundary, and comparisons all use that identical predicate. A higher target-priority weight alone cannot replace a mandatory requirement because a sufficiently high aggregate score can still omit that target.

The relation to DRL concerns value approximation and certification. A learned critic or actor could guide candidate ordering or propose actions, but a predicted value is not a completion certificate. Retaining the present guarantees would require checking the proposed action against a valid continuation and the relevant cost bound. This paper evaluates the model-based rollout instantiation and does not claim a trained deep policy, sample-efficiency result, or new DRL convergence theorem.

The experimental scope is deliberately explicit. The prescribed profiles are synthetic geometry-based tasks on one retained observation library. Their selection is independent of the new test outcomes but conditional on nominal observability in that library. They establish complete acceptance of specified samples under a deterministic camera model, not a validated set of mission-critical ISS components. Denser region sampling, verified component annotations, resolution and dwell acceptance, illumination, and independent geometry are needed for operational inspection claims. A fixed mandatory denominator must be retained when such requirements become difficult or unavailable.

Allowing safe zero-gain connections makes the declared graph action space consistent with directed reachability, at additional search cost. The finite-completion safeguard can still fail to find a route when its base policy is unsuccessful, even if a different route exists. Exact reduced-graph analysis and explicit unresolved-failure labels prevent that algorithmic limitation from being confused with physical infeasibility. The statistical comparisons also separate task completion from cost on the selected jointly successful subset.

Finally, HCW propagation, prescribed camera attitude, and the full-mesh numerical audits do not establish navigation robustness, actuator fidelity, continuous-space optimality, passive safety, image-based diagnosis, or flight performance. The historical ROS example verifies one corrected survey execution. The new target-credit checks prepare the execution interface for the revised task but do not substitute for a fresh ROS campaign.

VIII. Conclusion

This paper formulated complete prescribed-target inspection as viability-preserving rollout approximate dynamic programming over audited orbital arcs. A fixed required set defines mission success, while a finite Bellman prefix and complete adaptive base-policy value provide a checkable policy-improvement mechanism. Under the deterministic finite-graph assumptions, rollout preserves a finite completion certificate and, when the base policy completes, does not exceed that policy’s graph cost.

Across 50 new test and 30 shifted scenarios, ADP completed the required task in 43 and 9 cases, respectively. Matched-completion cost comparisons and the coverage-only ablation quantify the effects of sequence improvement and explicit task requirements separately. Every successful required-target route accepted all prescribed samples, while optional area coverage remained a distinct outcome. These results support the stated ADP formulation within the frozen synthetic task library; they do not establish full ISS inspection, trained DRL performance, or execution of the new task on flight hardware.

Data and Code Availability

The accompanying reproducibility package contains the corrected graph, fixed target profiles and their geometric provenance, configuration and freeze manifests, fresh scenario inventories, method–scenario rows, target-level credit records, statistical outputs, source tables, and figure scripts. Required-target evidence is versioned separately from the historical 80% benchmark, post-selection depth/coverage diagnostics, and accepted historical ROS task. A new required-target ROS bag is not included because that campaign was not run. The public NASA mesh source and checksum are documented; the mesh is not redistributed. A persistent repository identifier will replace the submission-package reference in the accepted version.

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